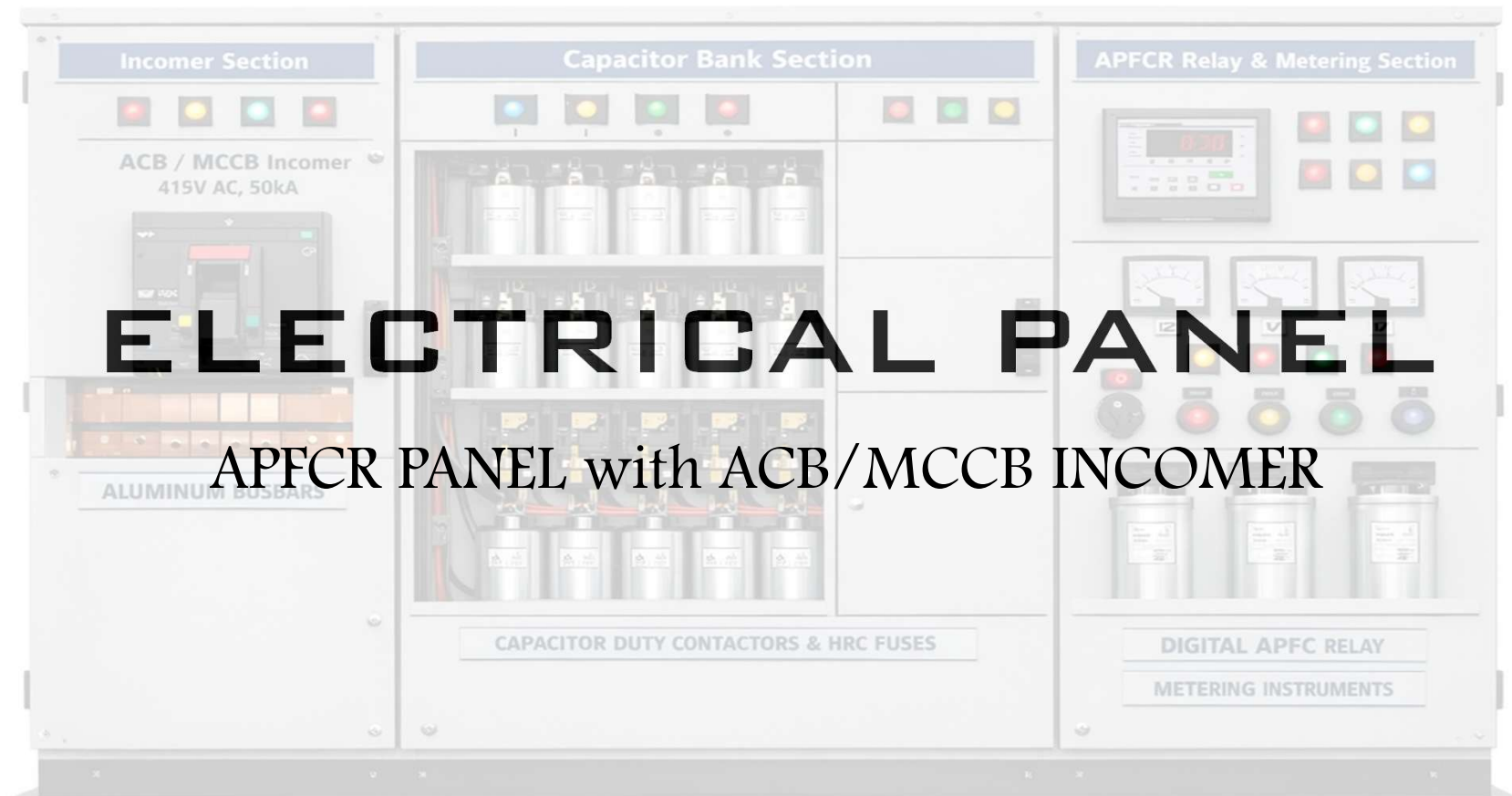


APFCR PANEL WITH ACB/MCCB INCOMER

Power Factor Improvement up to 0.99 Lag



ELECTRICAL PANEL

APFCR PANEL with ACB/MCCB INCOMER



IP54 PROTECTION



EARTHING
TERMINALS



FRLS
COPPER WIRING



APFCR PANEL WITH ACB/MCCB INCOMER

Power Factor Improvement up to 0.99 Lag

Description About APFCR PANEL

➤ **Design, Manufacturing, Supply, Installation, Testing & Commissioning (SITC) of Automatic Power Factor Correction Relay (APFCR) Panel** suitable for improving system power factor up to **0.99 lagging**, complete with **ACB/MCCB incomer**, capacitor banks, aluminum busbars, and all necessary accessories as per IS standards.

■ Panel Construction

- Incomer Section
- Busbar System
- Capacitor Bank
- APFCR Relay
- Capacitor Duty Contactors

- Protection System
- Metering and Indications
- Internal Wiring
- Earthing & Testing
- Accessories & Approved Makes (General)
- Documentation & Warranty

FRONT VIEW

REAR VIEW

LEFT VIEW

LEFT SIDE VIEW

1. PANEL CONSTRUCTION

- Floor mounted **cabicle type APFCR panel**, suitable for indoor installation.
- Fabricated from **CRCA sheet steel** (minimum 1.6 mm thick for structure and 1.6 mm for doors).
- Powder coated with **electrostatic epoxy powder coating** (minimum 70–80 microns thickness).
- Panel shall be **dust and vermin proof**, front access type.
- Degree of protection **IP54 or higher**.

2. INCOMER SECTION

- PANEL SHALL BE PROVIDED WITH SUITABLE RATED:

❖ ACB / MCCB INCOMER

- **AIR CIRCUIT BREAKER (ACB) / MOLDED CASE CIRCUIT BREAKER (MCCB) AS INCOMER.**
- **RATED VOLTAGE: 415V AC, 3 PHASE, 4 WIRE, 50 HZ.**
- **BREAKING CAPACITY: MINIMUM 50 KA OR AS REQUIRED.**
- **ADJUSTABLE THERMAL AND MAGNETIC PROTECTION.**
- **PROVIDED WITH SHUNT TRIP, UNDER VOLTAGE RELEASE, AUXILIARY CONTACTS IF REQUIRED.**

Floor mounted, CRCA Sheet steel, 1.6mm Thiup
Powder coatect tirio-uloasj powder costing(1000

Auxiliary contacts, Shunt Trip & Unvers Turflage
Auxiliary contacts, Shunt Trip & Auxiers Poweer Resenser)

3. BUSBAR SYSTEM

- **High conductivity Aluminum Busbars** with suitable current carrying capacity.
- Busbars supported on **SMC / DMC insulated supports**.
- Color coded **R-Y-B phases and Neutral**.
- Proper **heat shrinkable sleeves / PVC insulation** provided.
- Busbars designed to withstand **short circuit stresses**.

4. CAPACITOR BANK

- **HEAVY DUTY DETUNED / STANDARD APFC CAPACITORS SUITABLE FOR HARMONIC CONDITIONS.**

FEATURES :-

- ✓ 3 PHASE **APP TYPE OR MPP TYPE CAPACITORS**
- ✓ Rated voltage **440V / 525V**
- ✓ Provided with **capacitor duty contactors**
- ✓ Individual **HRC fuse protection**

- **TYPICAL STEP CONFIGURATION EXAMPLE :-**

- ✓ 5 KVAR
- ✓ 10 KVAR
- ✓ 20 KVAR
- ✓ 25 KVAR
- ✓ Total KVAR capacity as per site requirement.

- **Approved Makes :-**

- ✓ **Schneider**
- ✓ **EPCOS**
- ✓ **Ducati**
- ✓ **L&T**
- ✓ **Vishay**

5. APFCR RELAY

- Microprocessor based **Automatic Power Factor Control Relay** with :-
- Automatic capacitor switching
- LED/LCD display
- Programmable switching sequence
- Power factor measurement
- Protection against over-voltage / under-voltage

6. CAPACITOR DUTY CONTACTORS

- Special **capacitor switching contactors** with damping resistors.
- Suitable for frequent switching operations.

➤ **Approved Makes :-**

- ✓ L&T
- ✓ Schneider
- ✓ Siemens
- ✓ ABB

APFCR RELAY

➤ **Approved Makes :-**

- ✓ L&T
- ✓ Schneider
- ✓ Nippon
- ✓ Secure

CAPACITOR DUTY CONTACTORS

7. PROTECTION SYSTEM

➤ EACH CAPACITOR STEP SHALL BE PROTECTED WITH :-

- ✓ HRC Fuse / MCCB
- ✓ Capacitor duty contactor
- ✓ Indicating lamps for each step
- ✓ Over current protection

8. METERING AND INDICATIONS

➤ PANEL SHALL BE PROVIDED WITH :-

- ✓ Digital **Voltmeter** with selector switch
- ✓ Digital **Ammeter** with selector switch
- ✓ **Power Factor Meter**
- ✓ LED indication lamps for :-

- ☐ R phase
- ☐ Y phase
- ☐ B phase
- ☐ Capacitor steps

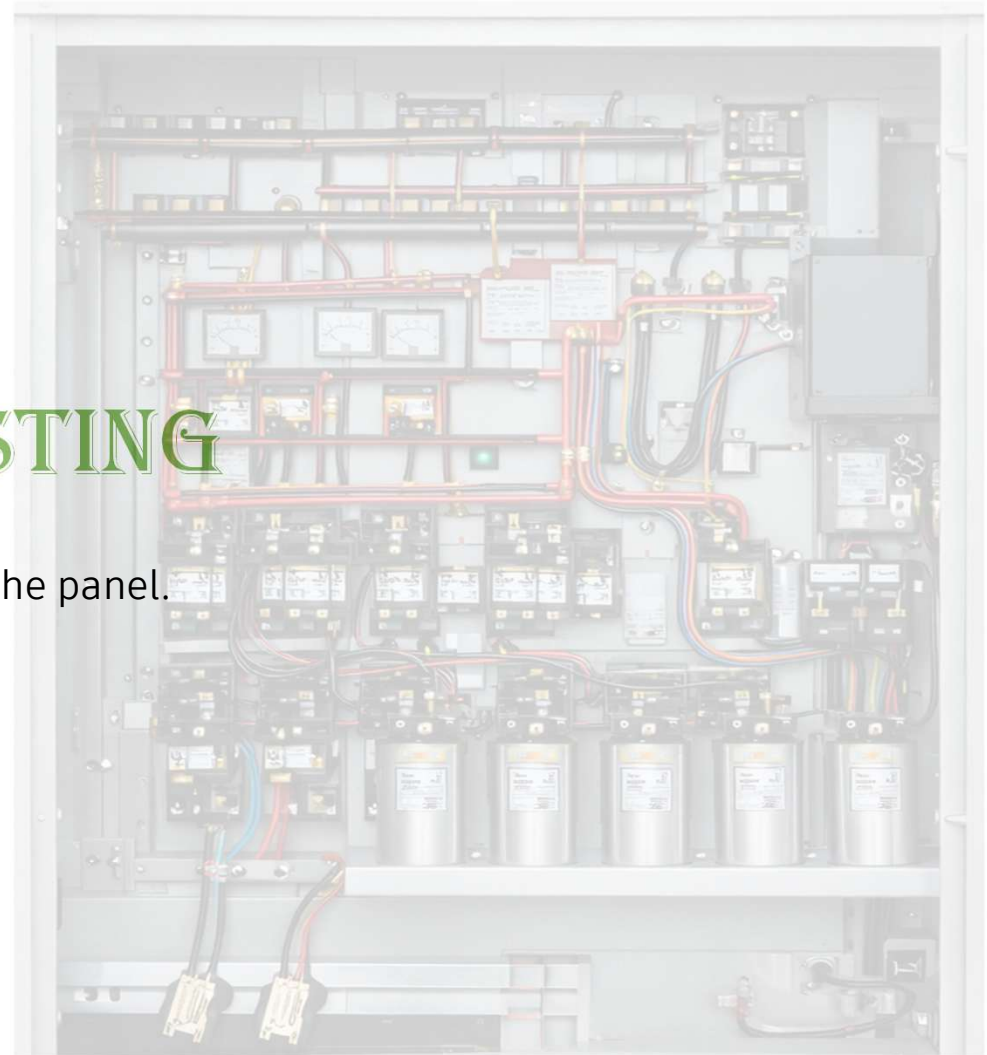
9. INTERNAL WIRING

- Wiring done with **FRLS copper wires**.
- Ferruled and properly numbered wiring.
- Neatly dressed wiring in **PVC channels**.

10. EARTHING & TESTING

- **EARTHING :-**
 - ✓ Two **separate earthing terminals** provided in the panel.
 - ✓ Earthing of panel body and door.
- **TESTING :-**
 - ❑ Panel shall be tested for:
 - ✓ Insulation resistance
 - ✓ Functional operation
 - ✓ Relay operation
 - ✓ Capacitor switching sequence
 - ❑ As per **IS 8623 / IEC 61439 standards**.

CABLE TERMINATION VIEW



INNER VIEW

11. ACCESSORIES & APPROVED MAKES (GENERAL)

➤ PANEL SHALL INCLUDE:

- ✓ Control MCB
- ✓ Terminal blocks
- ✓ Danger plate
- ✓ Name plate
- ✓ Cable entry glands
- ✓ Busbar insulation
- ✓ Internal lighting (if required)

➤ MAJOR COMPONENTS SHALL BE FROM APPROVED MAKES:

- ✓ L&T
- ✓ Schneider Electric
- ✓ Siemens
- ✓ ABB
- ✓ Havells
- ✓ Legrand

12. DOCUMENTATION & WARRANTY

➤ SUPPLIER SHALL PROVIDE :-

- ✓ Panel GA drawing
- ✓ Busbar layout
- ✓ Schematic diagram
- ✓ Test certificates
- ✓ Operation & maintenance manual

WARRANTY :-

Minimum **12 months warranty** from date of commissioning.